

1.1 Where is Alpha Truly Generated?

Traditionally, alpha is defined as the difference between the actual returns of an investment and the expected returns based on its risk profile. It is commonly expressed as:

$$[\alpha = R_i - (R_f + \beta (R_m - R_f))]$$

where

- R_i = return of the investment
- R_f = risk-free rate
- β = sensitivity of the investment to market risk
- R_m = return of the market

A positive alpha indicates skill in selecting securities and generating returns above what is expected given the investment's risk. Conversely, negative alpha suggests underperformance.

However, the nature of alpha has evolved significantly. The Efficient Market Hypothesis (EMH) argues that consistent alpha should not exist in a perfectly efficient market. Yet real markets exhibit inefficiencies arising from information frictions, structural constraints, and—critically—behavioural biases.

Technological progress has drastically reduced informational edge: data is widely accessible, trading is automated, and quantitative algorithms exploit micro-inefficiencies faster than any human could. As factor investing and smart-beta strategies commoditise previously profitable anomalies, genuine alpha increasingly requires *different* sources—those not easily captured by traditional risk models.

Behavioural biases fill that gap. Investors frequently deviate from rationality, especially under conditions of uncertainty, ambiguity, or limited attention. Collective emotional behaviour—fear, greed, overconfidence, herding—creates predictable price distortions. If these deviations from rationality are systematic and measurable, they can be modelled and harvested.

This evolution has blurred the line between alpha and beta. Many returns once attributed to “manager skill” are now understood as **alternative beta**—systematic factor exposures (value, momentum, quality, low volatility) that can be captured cheaply via rules-based portfolios. As these become commoditised, investors must identify *new* drivers of returns.

The focus of this research, therefore, is to identify **reliable behavioural indicators** that capture mispricings born from widespread cognitive errors or emotional reactions. These indicators form the foundation of a new class of systematic behavioural factors—what we refer to as *behavioural alpha*. As markets evolve, the most defensible sources of outperformance will increasingly depend on integrating unconventional data (sentiment, attention, ESG narratives) with behavioural finance frameworks.

1.2 Behavioural Finance as a Foundation for Alpha Generation

Behavioural finance posits that investors frequently rely on heuristics: mental shortcuts that evolved as efficient decision-making tools but can misfire under specific market conditions. These biases include **conservatism, overreaction, representativeness, loss aversion, anchoring, herding, and fear-driven selling**.

Importantly, heuristics do not always lead to errors. They only produce mispricing under certain conditions. Most low P/E stocks, for instance, are priced correctly. The challenge—and opportunity—is to identify when the wrong heuristic is applied at scale, generating systematic, predictable deviations from intrinsic value.

For behavioural mispricing to create exploitable alpha, two conditions must be met:

1. **Large groups of investors must exhibit the same bias, and**
2. **These biases must persist long enough for trading strategies to capture them.**

When both conditions hold, behavioural mispricings will emerge. They often tend to persist because arbitrage is limited by liquidity constraints, transaction costs, risk limits, institutional mandates, and career incentives. Thus, behavioural alpha is *endogenous*: it emerges from the structure and psychology of markets themselves.

By transforming behavioural biases into quantifiable indicators, we can construct factors that systematically capture behaviour-induced mispricings—bridging the gap between academic behavioural theory and practical portfolio construction.

1.3 Traditional and New Factors in Modern Investing

Traditional factors—value, size, momentum, low volatility, and quality—explain a substantial portion of cross-sectional return variation. In the last decade, however, new sources of return have emerged:

(1) Machine Learning and AI:

Models that identify nonlinear patterns in large datasets unreachable by traditional linear factor models.

(2) Alternative Data:

Unconventional datasets—satellite imagery, credit-card data, website traffic, social media sentiment—that provide new signals before they appear in financial statements.

(3) Behavioural Factors:

Signals derived from investor psychology, such as attention, sentiment, disposition, or herding.

(4) ESG/Narrative Factors:

Flows driven by narrative shifts—sustainability, climate transition, AI hype—affecting valuations independently of fundamentals.

The central research question of this paper is whether these behavioural and alternative data-driven indicators can be formalised into **factors** that consistently explain returns beyond traditional models.

YEARLY Q1-Q5 RETURN SPREAD %															
Frequency: Annually ▼															
Style	Factor	YTD▼	2024	2023	2022	2021	2020	2019	2018	2017	2016	2015	L3Y	L5Y	L10Y
Vol	90DCV	20.21	(4.54)	11.70	(4.04)	14.35	12.37	9.50	0.49	(11.27)	17.63	(10.27)	0.77	5.63	3.08
Vol	1MVol	17.09	(13.01)	23.74	(19.06)	6.75	3.35	11.03	(8.63)	(8.53)	7.19	(14.25)	(4.49)	(0.79)	(1.97)
Vol	12M Realized Volatility	14.46	(5.82)	22.03	(13.41)	7.91	16.90	0.29	(13.18)	(8.21)	8.76	(19.66)	(0.16)	4.65	(1.31)
HG	1Y Chg Sales Turnover	11.33	5.98	(2.09)	6.16	0.79	(19.27)	(3.43)	(8.98)	0.79	(11.17)	3.14	3.28	(2.17)	(3.13)
Vol	Volatility	11.14	(10.92)	28.10	(15.07)	11.38	11.35	7.39	(12.17)	(5.58)	12.32	(19.14)	(1.04)	3.75	(0.28)
HG	1Y Chg OCF / Assets	10.68	15.24	8.13	(8.81)	3.73	(10.78)	(10.58)	(0.41)	10.92	(10.44)	16.68	4.35	1.01	0.82
AE	Long Term Growth	10.12	10.93	16.68	(13.93)	(5.71)	9.87	5.24	(6.31)	16.26	(18.29)	9.72	3.67	2.91	1.72
PM	9M Price Momentum	10.05	19.35	(14.44)	8.78	(2.24)	(4.82)	(10.97)	12.06	6.46	(20.28)	26.02	3.57	0.66	0.98
AE	Analyst Expectations	8.98	12.11	(2.54)	(1.79)	5.41	(1.06)	(2.87)	3.02	15.47	(8.76)	10.10	2.38	2.28	2.65
HG	Historical Growth	8.64	9.22	1.63	(14.36)	1.36	(13.78)	(6.93)	2.50	4.72	(6.13)	5.68	(1.67)	(3.64)	(1.93)
AE	SUE	7.12	3.67	(1.18)	(2.83)	5.60	(1.89)	(2.28)	(0.84)	8.42	(2.16)	11.09	(0.15)	0.62	1.65
HG	1Y Chg EPS	6.71	11.22	(1.99)	(4.95)	(8.67)	(12.06)	(4.16)	(1.78)	4.12	(15.22)	9.06	1.19	(3.61)	(2.78)
HG	1Y Chg FCF / Assets	6.56	14.33	9.35	(4.58)	9.08	(4.19)	(11.89)	3.16	2.49	(1.66)	8.75	6.06	4.51	2.19
AE	Analyst Diffusion	4.80	9.20	(5.40)	7.63	7.74	(5.82)	1.54	5.83	11.38	(0.24)	3.51	3.59	2.44	3.38
Val	EBITDA / EV	3.50	0.50	(8.07)	23.26	8.29	(24.50)	(7.25)	(15.75)	3.25	13.12	(11.83)	4.43	(1.42)	(2.85)
EQ	WC Accruals	3.41	(1.48)	8.66	8.60	2.24	(1.26)	(3.15)	4.88	(1.72)	10.64	(8.50)	5.15	3.25	1.72
HG	Sustainable Growth	2.46	1.14	14.88	(15.50)	(6.97)	8.93	2.50	1.16	8.56	(6.78)	0.04	(0.61)	(0.10)	0.43
Val	Valuation	2.14	(2.12)	(10.27)	13.93	6.82	(17.39)	(2.86)	(10.01)	(3.51)	15.34	(11.62)	0.02	(2.46)	(2.72)
AE	Num FY1 EPS Rev	2.14	9.75	(4.30)	0.93	8.87	(9.86)	(3.76)	3.95	15.42	(0.27)	9.29	1.96	0.79	2.73
Val	Sales / EV	2.08	(2.27)	(3.68)	25.80	6.47	(8.97)	(4.61)	(5.98)	4.57	6.10	(10.10)	5.80	2.79	0.26
Val	E / P	1.94	1.94	(10.85)	7.58	1.08	(23.87)	(0.80)	(6.34)	5.48	7.48	(2.93)	(0.75)	(5.53)	(2.59)
Vol	60M CAPM Beta	1.50	0.31	17.63	(9.25)	12.40	11.76	7.90	(17.80)	6.81	10.19	(18.97)	2.31	6.11	1.30
CE	LT Debt to Equity	(0.78)	(5.31)	1.53	(7.51)	0.29	7.87	(5.12)	(4.73)	6.33	(4.27)	3.57	(3.84)	(0.77)	(0.87)
EQ	Net Profit Margin	(0.82)	6.88	(0.64)	(15.10)	1.59	(2.99)	(3.08)	1.07	10.18	(10.35)	6.82	(3.39)	(2.34)	(0.85)
Val	Div / Price	(0.97)	3.81	(17.35)	30.30	12.84	(18.54)	(2.27)	(4.86)	(4.59)	26.13	(5.23)	3.79	0.55	0.86
PM	12M - 1M Price Mom	(1.00)	19.63	(8.47)	11.89	(16.59)	(6.27)	(9.98)	2.92	9.20	(27.21)	29.19	7.00	(0.86)	(0.93)
CE	Return on Equity	(1.16)	9.37	4.37	(5.94)	(3.11)	(1.19)	0.73	2.70	7.69	(9.99)	7.10	2.40	0.55	0.99
CE	CF ROIC	(1.24)	10.39	9.39	(3.81)	5.82	8.86	(3.00)	5.59	15.62	(0.26)	6.91	5.12	6.00	5.38
Val	Book to Price	(2.56)	(2.76)	(9.40)	8.89	6.27	(17.36)	(3.42)	(17.15)	(9.32)	22.59	(17.80)	(1.38)	(3.37)	(4.73)
EQ	Accrual Ratio - CF	(4.76)	(7.16)	9.88	12.27	7.37	6.54	(3.27)	2.53	(11.35)	14.43	(1.14)	4.63	5.55	2.68

Returns
Heatmap:
●On○Off
 <-8.5
-8.5
-7.5
-6.5
-5.5
-4.5
-3.5
-2.5
-1.5
-0.5
0.5
1.5
2.5
3.5
4.5
5.5
6.5
7.5
8.5
>8.5

Source: Capital IQ

Up Market												
Style	Factor	Q1▼	Q5	Sprd%	Ann Sprd%	Q1 Hit	Q5 Hit	Avg IC	Best IC	Worst IC	IC T-Stat	%Pos IC
Vol	Volatility	5.06	2.37	2.69	37.51	84.04	85.46	0.09	0.66	(0.43)	6.70	68.44
Vol	12M Realized Volatility	4.94	2.39	2.55	35.27	84.40	86.88	0.08	0.61	(0.42)	6.60	67.38
Vol	60M CAPM Beta	4.89	2.38	2.51	34.58	85.82	85.11	0.09	0.64	(0.51)	7.59	68.79
Vol	1MVol	4.77	2.55	2.22	30.15	85.82	89.01	0.07	0.59	(0.42)	5.98	65.60
Vol	90DCV	4.72	2.75	1.97	26.44	85.46	91.13	0.06	0.60	(0.34)	5.55	62.77
AE	Long Term Growth	4.40	3.21	1.19	15.21	89.57	87.39	0.05	0.43	(0.39)	5.06	65.22
PM	5D Price Reversal	4.26	3.33	0.93	11.79	90.07	86.88	0.03	0.45	(0.30)	4.39	59.29
Sz	Log Market Cap	4.24	3.29	0.95	12.06	84.40	95.39	0.02	0.38	(0.32)	2.54	56.38
PM	1M Price Reversal	4.19	3.34	0.85	10.68	89.01	87.94	0.04	0.54	(0.39)	4.04	57.09
PM	1M Price High Low	4.14	2.85	1.30	16.70	90.43	90.07	0.05	0.46	(0.30)	6.19	63.12
Val	Book to Price	4.14	3.45	0.69	8.55	86.52	90.07	0.02	0.40	(0.42)	1.88	54.96
Val	FCF / Price	4.12	3.61	0.51	6.26	89.01	85.46	0.03	0.24	(0.29)	5.74	64.18
EQ	Accrual Ratio - CF	4.05	3.71	0.35	4.24	90.78	89.01	0.01	0.25	(0.19)	1.53	52.84
Val	EBITDA / EV	4.04	3.86	0.18	2.23	86.17	86.52	0.01	0.44	(0.35)	1.37	53.55
EQ	WC Accruals	3.96	3.65	0.31	3.82	91.49	90.43	0.00	0.21	(0.19)	0.58	49.29
Down Market												
Style	Factor	Q1▼	Q5	Sprd%	Ann Sprd%	Q1 Hit	Q5 Hit	Avg IC	Best IC	Worst IC	IC T-Stat	%Pos IC
EQ	Net Income Stability	(3.11)	(4.88)	1.77	23.40	15.75	11.64	0.07	1.00	(1.00)	3.37	65.75
Val	Div / Price	(3.23)	(5.08)	1.85	24.56	20.55	5.48	0.08	0.53	(0.36)	6.10	71.23
CE	CF ROIC	(3.37)	(4.81)	1.44	18.74	10.27	7.53	0.05	0.31	(0.24)	5.27	67.12
PM	9M Price Momentum	(3.39)	(5.56)	2.17	29.44	12.33	9.59	0.07	0.58	(0.45)	4.56	63.01
EQ	Earnings Quality	(3.39)	(4.53)	1.14	14.53	11.64	5.48	0.05	0.23	(0.17)	7.95	71.23
CE	Capital Efficiency	(3.41)	(4.81)	1.40	18.16	8.90	5.48	0.05	0.25	(0.20)	7.54	75.34
CE	Return on Equity	(3.47)	(4.88)	1.41	18.24	7.53	9.59	0.05	0.29	(0.33)	4.73	67.12
CE	1Y Chg Shares Out	(3.48)	(4.73)	1.25	16.11	14.38	4.79	0.04	0.34	(0.23)	4.86	65.75
PM	12M - 1M Price Mom	(3.50)	(5.58)	2.08	28.00	12.33	10.27	0.07	0.48	(0.41)	4.41	62.33
AE	Analyst Diffusion	(3.55)	(4.41)	0.86	10.77	12.39	6.19	0.04	0.28	(0.29)	4.06	68.14
AE	Num FY1 EPS Rev	(3.59)	(4.48)	0.89	11.23	9.73	10.62	0.04	0.37	(0.39)	3.60	65.49
PM	Price Momentum	(3.63)	(4.32)	0.69	8.62	11.64	6.16	0.02	0.48	(0.46)	1.91	57.53
Val	Valuation	(3.70)	(4.23)	0.52	6.46	13.70	5.48	0.01	0.43	(0.34)	1.13	51.37
EQ	Net Profit Margin	(3.72)	(4.90)	1.18	15.18	4.11	9.59	0.04	0.33	(0.35)	3.90	63.01
EQ	Cash Conversion Cycle	(3.83)	(4.19)	0.36	4.42	9.59	8.22	0.03	0.22	(0.14)	4.34	63.70

Source: Capital IQ

1.6 Hedge Funds as Proof-of-Concept: Behavioural Alpha in Practice

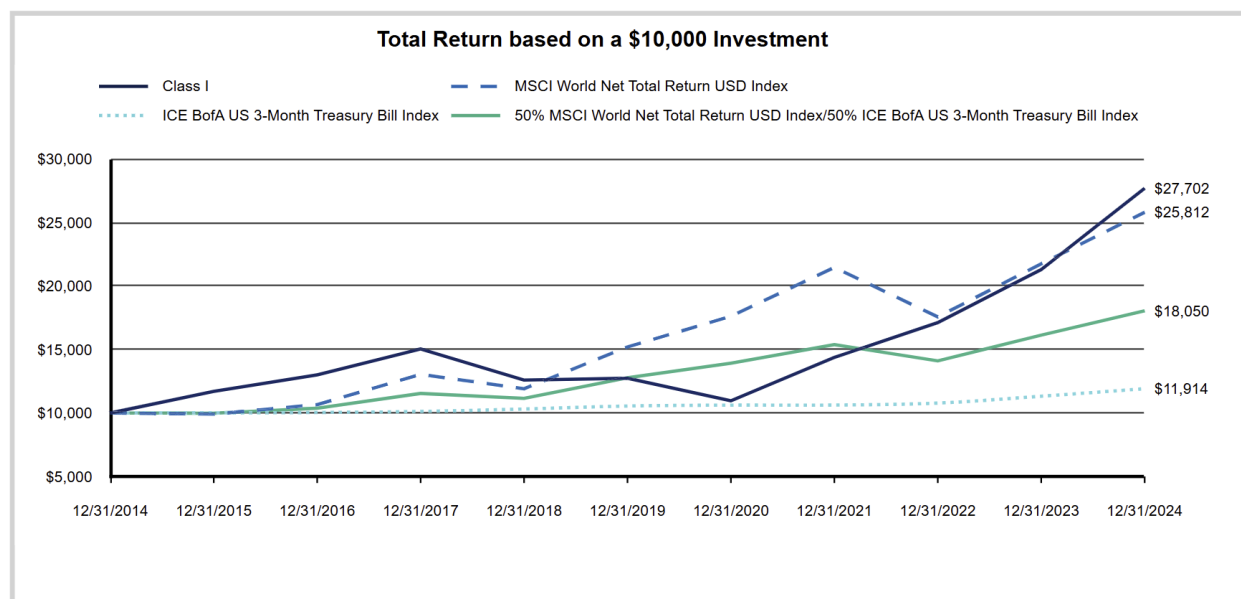
Behavioural alpha becomes credible not only through theory but through **institutional evidence**. The world's most sophisticated hedge funds—those with access to the best talent, datasets, computing power, and market microstructure intelligence—use systematic strategies designed precisely around investor biases.

A. Quantitative Systematic Hedge Funds

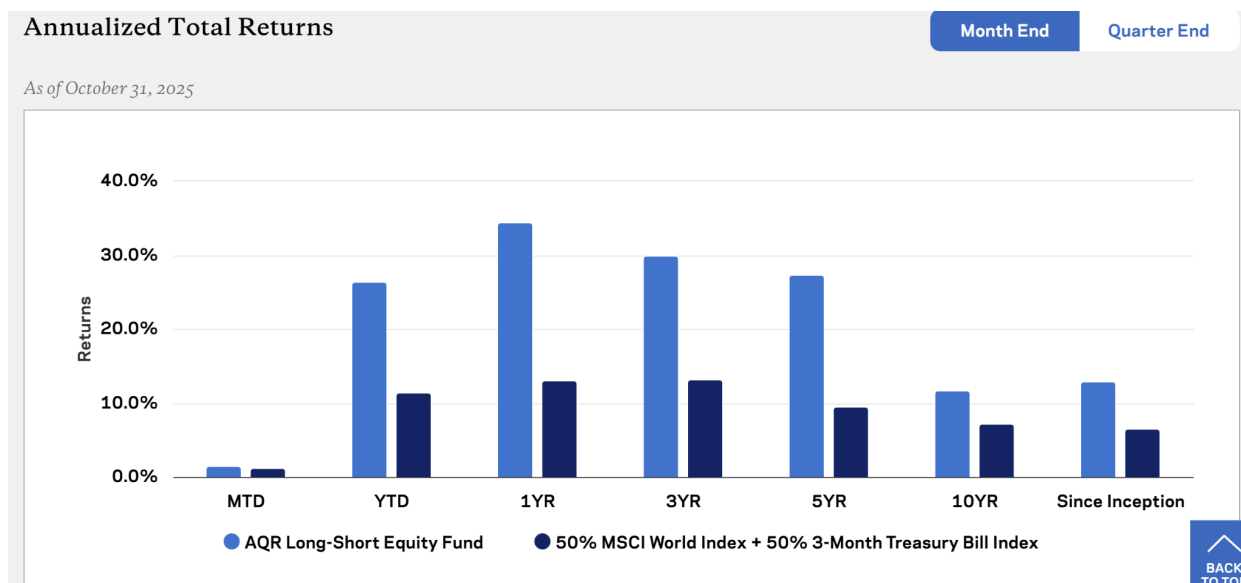
(1) AQR Capital Management Underreaction, Earnings Drift, and Slow Information Diffusion

Fund Background

AQR is one of the largest quantitative asset managers globally, combining academic finance with systematic trading. It is known for replicating and extending published anomalies into scalable strategies.



Source: AQR Website



Behavioural Alpha Insight

AQR's flagship anomalies are based on **slow information assimilation**, which occurs because investors struggle to parse complex earnings reports, analyst revisions follow a serially correlated and clustered pattern, institutional herding delays full market reaction AND liquidity providers

adjust slowly to new information.

Data Inputs

- earnings surprises (standardised and industry-adjusted)
- analyst revisions
- news sentiment measures
- institutional trading flows
- options-implied probabilities

Trade Structure

AQR constructs:

- **Long:** firms with strong positive earnings surprises + upward revisions
- **Short:** firms with negative surprises + downward revisions

Positions are sector- and factor-neutralised.

Why the Alpha Persists

- **Attention constraints** — investors underreact to complicated news.
- **Institutional frictions** — fund managers adjust positions gradually.
- **Career risk** — analysts revise incrementally rather than immediately.
- **Liquidity costs** — large institutions cannot rebalance instantly.

(2) Two Sigma — News Velocity, NLP-Based Sentiment, and Attention Overload

Fund Background

Two Sigma is a data-intensive hedge fund leveraging machine learning, natural language processing (NLP), and alternative data.

Behavioural Alpha Insight

Asset prices react not only to *what* is said but also *how intensely* information circulates. Two Sigma discovered:

- news velocity spikes produce immediate price pressure,
- but rapid sentiment shifts lead to reversals as attention dissipates.

Investors anchor on salient headlines and overweight fresh information.

Data Inputs

- linguistic sentiment (tone, polarity, subjectivity)
- novelty metrics (how different the news is from prior coverage)
- topic modelling (LDA, embeddings)
- news velocity/volume
- search trends (SVI)
- social sentiment

Trade Structure

- **Long:** high-news-velocity positive tone stocks, held 1–5 days
- **Short:** high-news-velocity negative tone stocks (oversold)
- **Reversal leg:** fade spikes after sentiment normalises

Why the Alpha Persists

- **Attention cycles** create temporary price dislocations.
- **Limited processing capacity** leads to overreaction to repeated news.
- **Noise traders** and retail investors amplify volatility.
- **Regulatory constraints** limit short-term arbitrage by institutions.